

Zach Taylor

971-832-4408 | Zacharyat19@gmail.com | [LinkedIn/zachtaylor22775](https://www.linkedin.com/in/zachtaylor22775) | [GitHub/Zacharyat19](https://github.com/Zacharyat19)

SUMMARY

OSU Computer Science graduate with experience developing performance focused software solutions for enterprise clients. Skilled in C++, Python, Next.js, and Docker with a focus on scalability, optimization, and the software development life cycle. Strong communication and collaboration skills, excelling in cross-functional teams.

EDUCATION

B.S. Computer Science - Oregon State University

September 2019 – June 2023

Won the Pioneering Concept Award for the most innovative project idea at the 2023 OSU Engineering Expo

PROJECTS

Patient Monitoring Robot | *Python, OpenCV, AI*

[GitHub](#)

- Spearheaded the initiative to enhance fall detection accuracy by **50%** on a team of **6** engineers, conducting live training in nursing homes and allowing caregivers to focus on other tasks while keeping patients safe.
- Presented project results to Intel sponsors and showcased progress at the 2023 OSU Engineering Expo, winning the Pioneering Concept Award for most innovative project idea.
- Developed a fall detection algorithm to address the lack of functionality in **OpenCV**, providing real-time updates that allow caregivers to respond immediately to patient falls.

SIMD Image Filter | *C++, SIMD*

[GitHub](#)

- Implemented Gaussian blur, Laplacian edge detection, and sharpen algorithms to demonstrate the benefits of SIMD, achieving **5x–7x** faster performance compared to scalar versions.

EXPERIENCE

Software Engineer Intern *Rohde & Schwarz*

April 2024 – Jan 2025

- Implemented **C++** solutions for testing Cellular-Vehicle-to-Everything (C-V2X) networks, providing customers with detailed network diagnostics to identify performance bottlenecks and support safer vehicle communication.
- Designed a composite **API** to replace outdated and fragmented legacy code, simplifying integration into future software products and achieving a **3x–4x** speed increase in benchmark tests.
- Developed **unit tests** for new APIs, uncovering **15** hidden computation errors across all products; resolved issues with a patch, reducing bug reports and streamlining testing procedures for future releases.
- Automated **SSH** and Ethernet configuration for team members with **Python** scripting, reducing development and testing time by **8%** across multiple sprints.

Software Development Intern *Portland State University*

June 2019 – September 2019

- Designed a **TensorFlow** neural network model with a team of **4** engineers using **Python** and **OpenCV** for food processing efficiency and automating quality control on assembly lines.
- Enhanced the model using transfer learning to speed up development and improve object detection accuracy, achieving **97%** detection in production tests.
- Coordinated weekly meetings with engineers and end users to ensure alignment between product development and operational goals, keeping the project on track with deadlines.

Undergraduate Teaching Assistant *Oregon State University*

September 2020 – June 2023

- Delivered curriculum in **C** for data structures and operating systems courses, with supplemental material designed to help students build a foundation on key language features where formal instruction was missing.
- Advised students on end-of-term projects for the **Full-Stack** development course, providing guidance on design decisions and helping them translate ideas into effective applications.
- Automated grading process using **Linux** scripting, reducing grading time and streamlining workflow to better balance work and school.
- Proctored exams of **100–150** students to prevent misconduct, upholding academic integrity and maintaining a fair, distraction-free testing environment.

SKILLS

C++, Python, Typescript, JavaScript, Next.js, TensorFlow, CUDA, GitHub, Linux, Jenkins, AWS, Docker